

PAGE 1: KHARIF SEASON AGRONOMY & CROPPING CALENDAR

Definition & Overview: Kharif season, also known as the monsoon cropping season, begins with the onset of the southwest monsoon in June.

Key Kharif Crops: Rice (Paddy), Maize (Corn), Cotton, Soybean, Groundnut, Pigeonpea

Sowing Windows & Monsoon Dynamics:

1. North India (Punjab, Haryana, UP): Sowing window is June 15 to July 15. For transplanted rice, the window is narrower, typically June 15 to June 30.
2. Central & Western India (MP, Maharashtra, Gujarat): Sowing commences with the first monsoon rains in June and continues through July.
3. Eastern & Southern India: Sowing spans June 1 to July 31 depending on monsoon advancement.

Critical Agronomic Management:

- Land Preparation: Deep summer ploughing followed by 2 cross-harrowings to destroy weeds and level the field.
- Drainage Planning: Provide surface drainage channels (slope 0.1-0.2%) to prevent waterlogging.
- Seed Priming & Fungicide Inoculation: Treat all kharif seeds with Trichoderma viride @ 5g/kg.

PAGE 2: RABI SEASON AGRONOMY & WINTER CROPPING

Definition & Dynamics: Rabi cropping season spans winter and spring, sown from October to

Major Rabi Crops: Wheat, Rapeseed & Mustard, Chickpea \ (Bengal Gram\), Field Pea, Lentil

Temperature Requirements & Vernalization:

- Germination Temperature: 20°C - 25°C.
- Tillering & Vegetative Stage: Cool weather with temperatures between 10°C - 15°C promoti
- Grain Filling & Ripening: Warm, clear sunny days \ (20°C - 28°C\). Avoid terminal heat

Agronomic Interventions:

- Timely Sowing: Sowing wheat between November 1 and November 20 maximizes yield potential
- Crown Root Initiation \ (CRI\) Irrigation: First irrigation must be applied precisely a
- Pre-sowing Seed Treatment: Treat chickpea and pulse seeds with Rhizobium culture and Pho

PAGE 3: ZAID \((SUMMER)\) SEASON CROPPING & HEAT ADAPTATION

Definition & Opportunity: Zaid season is the short summer cropping window between Rabi har

Major Zaid Crops: Summer Moong \((Green Gram - SML 668, IPM 205-7, Samrat)\), Summer Urad

Agronomic Management Protocols:

1. Rapid Field Turnaround: Sow summer moong immediately after wheat or potato harvest by M

PAGE 4: AGRO-CLIMATIC ZONES OF INDIA & REGIONAL PLANNING

The Planning Commission delineated 15 distinct Agro-Climatic Zones based on physiography,

1. Western Himalayan Region \ (J&K, HP, Uttarakhand\): Apple, saffron, walnut, temperate
2. Eastern Himalayan Region \ (Assam, NE States, Sub-Himalayan WB\): Tea, rice, pineapple
3. Lower Gangetic Plains \ (West Bengal\): Rice-rice and rice-jute cropping systems, fish
4. Middle Gangetic Plains \ (UP, Bihar\): Rice-wheat, maize, sugarcane, pigeonpea, mustard
5. Upper Gangetic Plains \ (Western UP\): High-intensity rice-wheat, sugarcane, potato, m
6. Trans-Gangetic Plains \ (Punjab, Haryana, Delhi, Rajasthan Ganganagar\): Highly mechan
7. Eastern Plateau & Hills \ (Chota Nagpur, Odisha, MP\): Rainfed rice, minor millets, ni
8. Central Plateau & Hills \ (Bundelkhand, Malwa, MP\): Soybean, wheat, chickpea, mustard
9. Western Plateau & Hills \ (Maharashtra\): Cotton, sugarcane, jowar, bajra, pomegranate
10. Southern Plateau & Hills \ (AP, Telangana, Karnataka, TN\): Groundnut, cotton, ragi,
11. East Coast Plains \ (Coastal AP, TN, Odisha\): Coastal paddy, coconut, cashew, oil pa
12. West Coast Plains & Ghats \ (Kerala, Coastal Karnataka, Goa\): Spices, rubber, coconu
13. Gujarat Plains & Hills: Cotton, groundnut, castor, cumin, sesame, bajra.
14. Western Dry Region \ (Western Rajasthan\): Pearl millet, cluster bean \ (guar\), mot
15. Island Region \ (Andaman & Nicobar, Lakshadweep\): Coconut, arecanut, tropical fruits

PAGE 5: SCIENTIFIC CROP SELECTION FRAMEWORK

The 6-Pillar Decision Framework for Crop & Variety Selection:

Pillar 1: Soil Characteristics:

- Heavy Clay Soils \(\Vertisols\): High water retention; ideal for Cotton, Soybean, Wheat
- Sandy Loam Soils \(\Inceptisols/Entisols\): Excellent aeration; ideal for Potato, Groun
- Saline/Alkaline Soils: Choose tolerant crops \(\Barley, Mustard, Cotton, Beetroot\).

Pillar 2: Water Availability & Irrigation Source:

PAGE 6: CLIMATE CHANGE ADAPTATION & NICRA TECHNOLOGIES

National Innovations in Climate Resilient Agriculture \NICRA\ Strategic Interventions:

1. Flash-Flood & Submergence Adaptation:

- Deploy Swarna-Sub1, Samba Mahsuri-Sub1, and CR Dhan 801 rice varieties capable of surviv

2. Drought Adaptation & Moisture Scarcity:

- Use Sahbhagi Dhan, CR Dhan 201, and DRR Dhan 42 for direct dry seeding, maturing in 110-

3. Terminal Heat Stress in Wheat:

PAGE 7: CROP DIVERSIFICATION & INTERCROPPING RATIOS

Intercropping maximizes Land Equivalent Ratio ($LER > 1.25$), reduces economic risk, and Standard Recommended Intercropping Combinations & Spatial Geometries:

1. Wheat + Mustard (8:1 or 9:1 row ratio):

- 8 rows of Wheat (PBW 824 / HD 3086) spaced at 20 cm alternating with 1 row of Mustard

2. Chickpea + Mustard (6:1 or 4:1 row ratio):

- Chickpea (JG 14) fixes atmospheric nitrogen, reducing mustard fertilizer needs.

3. Maize + Cowpea / Soybean (1:1 or 2:2 row ratio):

- Provides balanced cereal-legume fodder and grain while smothering weeds.

4. Sugarcane + Potato / Onion / Garlic (1:2 row ratio):

- Short-duration crops grown in inter-row space (90-120 cm) of autumn sugarcane provide

5. Cotton + Green Gram / Black Gram (1:2 row ratio):

- Legumes cover the ground during initial 60 days of slow cotton growth, reducing weed emergence

PAGE 8: SUSTAINABLE CROP ROTATION CYCLES

Monoculture (continuous Rice-Wheat or Cotton-Cotton) leads to groundwater depletion, m

Recommended Sustainable Rotational Cycles:

Cycle 1: Rice - Wheat - Summer Moong (3 crops/year):

- Green gram incorporated into soil after 2 pod pickings adds 35-40 kg N/ha and breaks the

Cycle 2: Cotton - Wheat - Fallow (2-year cycle):

- Replaced with Cotton - Chickpea or Cotton - Mustard to conserve 40% winter irrigation.

Cycle 3: Maize - Mustard - Summer Moong (Intensive Low-Water Rotation):

- Consumes 60% less water than Rice-Wheat while providing equivalent net returns.

Cycle 4: Soybean - Wheat - Green Manuring (Dhaincha):

- Central India rotation optimizing soil organic carbon and reducing urea requirement by 2

Rotational Benefits:

- Reduces nematode populations by 70% when non-host crops (Mustard, Marigold) are rota
- Destroys Phalaris minor seed bank in wheat when replaced with Berseem or Sunflower.

PAGE 9: NATURAL FARMING & BIO-INPUT PREPARATION

Zero Budget Natural Farming (ZBNF) is based on 4 Core Pillars:

1. Jeevamrutha (Liquid Microbial Bio-Enhancer):

- Recipe: 200 L water + 10 kg fresh indigenous desi cow dung + 5-10 L desi cow urine + 2 kg
- Fermentation: Stir clockwise twice daily in shade for 48-72 hours.
- Application: Apply 200 L/acre through irrigation water or as 10% foliar spray every 15-20 days

2. Beejamrutha (Seed Microbial Shield):

- Recipe: 20 L water + 5 kg desi cow dung + 5 L cow urine + 50 g slaked lime (chuna) + 50 g
- Application: Slurry coat on seeds before sowing; prevents seed-borne and soil-borne fungi

3. Achhadana (Mulching):

- Soil mulching with dry crop residue (3-4 tonnes/acre) keeps root zone cool, conserve

4. Whapasa (Soil Moisture-Aeration Equilibrium):

- Water is applied only to alternate furrows during afternoon, creating vapor aeration rate

PAGE 10: AGROFORESTRY & BOUNDARY PLANTATION

Agroforestry integrates woody perennials with agricultural crops on the same land manageme

Agroforestry Models:

1. Poplar \(\textit{Populus deltoides}\) + Agri-Crops \(\text{North India}\):

- Spacing: 5 m x 4 m or 7 m x 3 m \((500 \text{ trees/ha})\).
- Compatible Crops: Sugarcane, Wheat, Turmeric, Ginger, Potato during initial 3 years. Dec
- Economics: 5-year timber harvesting generates Rs 4,00,000 - 6,00,000/acre net timber rev

2. Melia dubia \(\text{Malabar Neem}\) + Pulses/Vegetables \(\text{South \& Central India}\):

- Fast-growing plywood tree harvesting in 6-7 years. Tolerates high temperature and provid

3. Windbreak & Shelterbelt Design:

- Plant 2-3 staggered rows of Casuarina, Bamboo, or Shisham along the south-western bounda
- Reduces wind velocity by 50-60%, slashing crop evapotranspiration loss by 20-25% and pre

PAGE 11: SOIL TESTING METHODOLOGY & SOIL HEALTH CARD

Soil testing is the scientific assessment of soil nutrient availability, chemical balance, Representative Soil Sampling Protocol:

1. Sampling Pattern: Divide farm into uniform management units based on slope, color, and
2. Sampling Depth: 0-15 cm for field crops \ (cereals, pulses, oilseeds\); 0-30 cm for de
3. Sample Processing: Mix sub-samples thoroughly, quarter down to 500 grams, shade dry, cr

The 12 Critical Soil Health Card Parameters:

PAGE 12: SOIL pH & ACID SOIL RECLAMATION

Soil Reaction (pH) Dynamics: Soil pH measures hydrogen ion activity. Optimal nutrient

Acid Soil Constraints (pH < 6.0):

- Aluminum (Al^{3+}) and Manganese (Mn^{2+}) toxicity causes root stunting and 'stubby r
- Phosphorus gets chemically fixed as insoluble Aluminum and Iron Phosphates, resulting in
- Beneficial bacterial populations (Rhizobium, Nitrosomonas) drop by up to 80%.

Reclamation & Liming Protocols:

PAGE 14: SOIL ORGANIC CARBON (SOC) ENHANCEMENT

Soil Organic Carbon (SOC) is the fundamental indicator of soil biological vitality, ca

Proven SOC Enhancement Strategies:

1. Farmyard Manure (FYM) & Well-Decomposed Compost:

- Apply 8-10 tonnes/ha well-rotted FYM (C:N ratio 20:1 to 25:1) 3 weeks before sowing.

2. Vermicomposting (Eisenia fetida Earthworms):

- Apply 2.5 - 3.5 tonnes/ha vermicompost. Contains 5x available N, 7x available P, 11x ava

3. In-situ Green Manuring:

- Sow Dhaincha (Sesbania aculeata) or Sunnhemp (Crotalaria juncea) @ 50 kg/ha with
- Plough down at 45-50 days (flowering stage). Adds 20-25 tonnes/ha succulent green bi

4. Retention of Crop Residues (Zero-Burn Policy):

- Retaining wheat/paddy stubble (5-7 tonnes/ha) using Super Seeder or Happy Seeder pre

PAGE 15: SOIL COMPACTION & SUBSOILING

Soil Physical Matrix & Compaction Hazards:

- Ideal Soil Composition: 45% mineral matter, 5% organic matter, 25% soil water, 25% soil
- Subsoil Hardpan Formation: Continuous rotary tillage at 10-15 cm and repeated heavy trac
- Consequences: Root penetration stops, water infiltration drops by 75%, causing waterlogg

Diagnostic & Remediation Protocol:

1. Penetrometer Diagnosis: A cone penetrometer resistance exceeding 2.0 MPa \ (300 PSI)
2. Subsoiling \ (Chisel Ploughing)\:
 - Equipment: Single or multi-shank subsoiler penetrating 45-60 cm depth.
 - Spacing: Run subsoiler at 1.0 to 1.5 meter spacing across the field in dry summer condit
 - Frequency: Once every 3 to 4 years.
3. Yield Impact: Shattering hardpan increases cotton root depth from 30 cm to 90 cm, boost

PAGE 16: LASER LAND LEVELLING & SEEDBED ENGINEERING

Laser Land Levelling (LLL) Technology:

- Working Principle: A tractor-mounted scraper guided by a rotary laser transmitter and receiver.
Quantified Agronomic & Economic Benefits:

1. Water Savings: Reduces irrigation water requirement by 25-30% due to rapid, uniform water distribution.
2. Cultivable Area Increase: Eliminates unnecessary internal bunds and irrigation channels.
3. Fertilizer Efficiency: Enhances fertilizer use efficiency by 15-20% by preventing nutrient loss.
4. Yield Boost: Produces uniform germination, synchronous tillering, and an average yield increase of 10-15%.
5. Service Life: A single laser levelling remains effective for 3-4 years under standard tillage conditions.

PAGE 17: CONSERVATION AGRICULTURE & DIRECT SEEDED RICE (DSR)

Conservation Agriculture (CA) is defined by 3 Core Principles: (1) Minimum mechanization
Zero-Till Wheat Sowing:

- Direct sowing of wheat immediately after paddy combine harvest without any field preparation
- Saves Rs 2,500 - 3,500/ha in diesel tillage costs and allows 10-12 days earlier sowing,

Direct Seeded Rice (DSR) Technology (Tar-Water Method):

1. Land Preparation: Laser level field, apply pre-sowing heavy irrigation (Rauni), and
2. Sowing: Drill primed seed @ 20-25 kg/ha at 3-4 cm depth with DSR machine equipped with
3. Weed Management (Critical):
 - Spray Stomp 30% EC (Pendimethalin) @ 1.0 L/acre within 24 hours of sowing.
 - Follow with Nominee Gold (Bispyribac-sodium 10% SC) @ 100 ml/acre or Almix @ 8 g/acre
4. Water Saving: Conserves 20-25% groundwater by eliminating continuous puddling and ponding

PAGE 18: SOIL MICROBIOLOGY & BIOFERTILIZER SCIENCE

Beneficial soil microorganisms convert unavailable soil mineral reserves into plant-absorb

1. Nitrogen-Fixing Biofertilizers:

- Rhizobium spp.: Symbiotic N-fixer for legumes \ (Chickpea, Soybean, Moong, Groundnut).

PAGE 19: SOIL MICRONUTRIENT DEFICIENCY & REMEDIATION

Intensive agriculture has led to widespread multi-micronutrient deficiencies across 45% of

Specific Micronutrient Diagnostics & Soil/Foliar Remedies:

1. Zinc (Zn) Deficiency (Khaira Disease in Rice / White Bud in Maize):

- Symptoms: Rusty brown pigmentation on middle leaves, stunted internodes, delayed maturity
- Soil Application: Apply Zinc Sulfate Heptahydrate (21% Zn) @ 25 kg/ha or Zinc Sulfate
- Foliar Emergency: Spray 0.5% Zinc Sulfate (5 g/L) + 0.25% Lime (2.5 g/L) or 0.1%

2. Iron (Fe) Chlorosis:

- Symptoms: Interveinal chlorosis on youngest leaves turning completely bleached white. Co
- Foliar Remedy: Spray 1.0% Ferrous Sulfate (FeSO₄ 19% Fe @ 10 g/L) + 0.1% Citric Acid

3. Boron (B) Deficiency (Hollow Heart in Groundnut / Cracking in Fruit):

- Soil: Apply Borax (10.5% B) @ 10 kg/ha or Solubor (20% B) @ 5 kg/ha basal.
- Foliar: Spray 0.1-0.15% Solubor (1.0-1.5 g/L) at flower pre-bloom.

4. Sulfur (S) Deficiency (Yellowing of Young Leaves in Mustard/Pulses):

- Soil: Apply Agricultural Gypsum @ 250 kg/ha or Bentonite Sulfur 90% @ 25 kg/ha at sowing

PAGE 20: SOIL & WATER CONSERVATION ENGINEERING

Soil erosion strips topsoil rich in organic matter and nutrients, causing land degradation

Engineering & Agronomic Soil Conservation Structures:

1. Contour Bunding & Graded Bunding:

- Suitable for slopes between 2% and 6%. Earthen embankments constructed along elevation c
- Impounds runoff, allowing 80% rainwater percolation and reducing soil loss from 25 t/ha/

2. Bench Terracing:

- Essential on steep slopes (>15% in hilly states). Converts sloping land into a serie

3. Vegetative Barriers (Vetiver Grass - *Vetiveria zizanioides*):

- Planting dense hedge rows of Vetiver grass on contours at 1-meter vertical drops creates

4. Gully Plugging & Check Dams:

- Construct loose boulder check dams and gabion structures across drainage gullies to redu

PAGE 21: PRECISION AGRICULTURE & SMART FARMING PILLARS

Definition & Core Principles: Precision Agriculture \Smart Farming\ is a data-driven f

Key Pillars & Technologies:

1. Global Navigation Satellite Systems \GNSS\ & RTK Auto-Steer: Centimeter-level posit
2. Variable Rate Technology \VRT\): Electro-hydraulic control of seed and fertilizer ap
3. IoT Sensor Telemetry: Soil moisture, EC, and leaf temperature sensors transmitting fiel
4. Drone Multispectral Imaging: High-resolution NDVI monitoring for stress detection 7-10
5. Edge AI Agronomy Decision Engines: Predictive algorithms calculating daily crop water b

PAGE 22: AUTOMATED FERTIGATION & WATER-SOLUBLE FERTILIZERS

Definition & Overview: Fertigation is the precise injection of 100% water-soluble fertiliz

Compatible Water-Soluble Formulations:

- 19:19:19 \ (Starter/Vegetative\)
- 12:61:0 \ (Mono Ammonium Phosphate MAP - Root initiation & flowering\)
- 13:0:45 \ (Potassium Nitrate - Fruit enlargement & grain filling\)
- 0:52:34 \ (Mono Potassium Phosphate MKP - Reproductive maturity & disease resistance\)
- 0:0:50+17.5%S \ (Sulfate of Potash SOP - Quality, brix sugar, and color\).

Fertigation Compatibility Rules:

PAGE 24: ALTERNATE WETTING AND DRYING (AWD) WATER PROTOCOL

AWD Protocol Details: Developed by IRRI, AWD is an efficient irrigation strategy where the
Implementation Steps:

1. Install a 'Pani Pipe' (30 cm long, 10-15 cm diameter perforated plastic pipe) in the
2. After transplanting, keep 2-5 cm water for 2 weeks to establish roots.
3. Thereafter, allow water to subside. When water level in pipe drops to 15 cm below soil

PAGE 25: CRITICAL CROP GROWTH STAGES FOR IRRIGATION

Moisture stress at critical phenological stages causes irreversible yield loss:

1. Wheat: (a) Crown Root Initiation CRI (21 DAS - most critical), (b) Tillering
2. Rice: (a) Seedling establishment, (b) Active Tillering, (c) Panicle Initiation
3. Maize: (a) Tasseling (VT) and (b) Silking (R1). Stress at silking causes
4. Cotton: (a) Squaring, (b) Peak Flowering, (c) Early Boll Development.
5. Chickpea: (a) Pre-flowering branching, (b) Pod development. DO NOT irrigate at

PAGE 26: SUB-SURFACE DRIP IRRIGATION \ (SDI\)

SDI Architecture: Drip laterals embedded 15 to 30 cm below soil surface directly in the cr

Key Advantages:

- Zero evaporation loss from soil surface.
- Zero weed germination between crop rows as topsoil remains dry.
- Allows uninterrupted tractor and equipment movement on dry surface.
- Reduces lateral degradation from UV radiation and rodent damage.
- Delivers 95% water and nutrient application efficiency.

Technical Requirements: Heavy-duty pressure compensating \ (PC\) emitters with anti-sipho

PAGE 27: IoT SOIL MOISTURE SENSING & AUTOMATION

Sensor Technologies:

1. Frequency Domain Reflectometry (FDR) & Capacitance Probes: Measure dielectric permittivity
2. Granular Matrix Sensors (Watermark): Measure soil water tension in centibars/kPa.

Irrigation Trigger Thresholds:

PAGE 28: PROTECTED CULTIVATION & HYDROPONIC SYSTEMS

Protected Cultivation Structures:

1. Naturally Ventilated Polyhouse (NVPH): G.I. pipe frame covered with 200-micron UV-s
2. Climate-Controlled Greenhouse: Evaporative cooling pad-and-fan system with foggers, kee

Hydroponic Soilless Systems:

- Nutrient Film Technique (NFT): Thin film of recirculating nutrient solution over pla
- Dutch Bucket System: Bato buckets filled with perlite/cocopeat substrate with automated
- Water Savings: Uses 90% less water than traditional open-field farming.

PAGE 29: SOLAR IRRIGATION & PM-KUSUM SCHEME

Solar Pumping Architecture: Photovoltaic (PV) array (3 HP to 10 HP) connected to a PM-KUSUM Government Scheme Components:

- Component A: Decentralized solar power plants (0.5 to 2 MW) on barren farmlands.

PAGE 30: POST-HARVEST STORAGE & COLD CHAIN

Post-Harvest Loss Mitigation Technologies:

1. Grain Hermetic Storage \PICS Bags / Cocoons\): Multi-layer polyethylene hermetic bag
2. IoT-Monitored Grain Silos: Temperature and humidity sensor cables inside galvanized cor
3. Solar-Powered Micro Cold Rooms: 5-10 MT cold rooms operating on thermal energy storage

PAGE 31: BENEFITS OF MODERN FARMING & AGRITECH ADOPTION

Key Benefits of Modern Farming Technologies:

1. Resource Efficiency: Drip irrigation saves 40-60% water while automated fertigation red
2. Yield Optimization: Precision farming tools and hybrid seed selection yield 20-35% high
3. Labor & Time Savings: Drone UAV spraying completes 1 acre in 7-10 minutes compared to 3

PAGE 32: SOIL HEALTH CARD PARAMETERS & AMELIORATION

Soil Health Card 12 Key Indicators & Chemistry Benchmarks:

1. pH (1:2.5 soil-water): Strongly Acidic (<5.5), Slightly Acidic (5.5-6.5), N

PAGE 33: PLANT NUTRIENT DYNAMICS & ADVANCED NUTRITION

Core Nutrient Principles: Plant nutrition requires balanced availability of 17 essential e

Fertilizer Dosage & Placement:

- Basal Application: Apply 100% of Phosphorus and Potassium alongside 25-33% Nitrogen at s
- Top Dressing: Apply remaining Nitrogen in 2-3 equal splits at active tillering/branching
- Foliar Supplementation: Spray 1% 19:19:19 or 0.5% chelated zinc during vegetative growth
- Soil Health Integration: Combine inorganic fertilizers with 5-10 tonnes/ha FYM to enhanc

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PAGE 41: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS

Phenological Milestone: Seed Germination & Emergence

Agronomic Characterization:

- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 42: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS

Phenological Milestone: Active Vegetative & Tillering

Agronomic Characterization:

- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 43: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS

Phenological Milestone: Stem Elongation & Jointing

Agronomic Characterization:

- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 44: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS

Phenological Milestone: Panicle Initiation & Booting

Agronomic Characterization:

- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 45: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS

Phenological Milestone: Flowering & Anthesis

Agronomic Characterization:

- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 46: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS

Phenological Milestone: Fruit Set & Pod Development

Agronomic Characterization:

- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 47: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS

Phenological Milestone: Grain Filling & Milk Stage

Agronomic Characterization:

- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 48: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS

Phenological Milestone: Dough & Maturity Stage

Agronomic Characterization:

- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 49: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS

Phenological Milestone: Harvesting Indices & Threshing

Agronomic Characterization:

- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 50: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS

Phenological Milestone: Post-Harvest Curing & Moisture

Agronomic Characterization:

- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 51: INTEGRATED WEED MANAGEMENT (IWM) & HERBICIDE ROTATION

Weed Control Protocols:

1. Cultural Control: Stale seedbed technique, smother intercropping, and high-density sowing
2. Mechanical Control: Cono-weeding in wet rice at 15 and 30 DAT; wheel hoeing in upland rice
3. Chemical Control:
 - Pre-Emergence Herbicides: Apply Pendimethalin 30% EC @ 1.0 L/acre or Pretilachlor 50% EC
 - Post-Emergence Herbicides: Apply Bispyribac-sodium 10% SC @ 100 ml/acre in rice or Clodinafop-methyl 15% EC

PAGE 52: INTEGRATED WEED MANAGEMENT (IWM) & HERBICIDE ROTATION

Weed Control Protocols:

1. Cultural Control: Stale seedbed technique, smother intercropping, and high-density sowing
2. Mechanical Control: Cono-weeding in wet rice at 15 and 30 DAT; wheel hoeing in upland rice
3. Chemical Control:
 - Pre-Emergence Herbicides: Apply Pendimethalin 30% EC @ 1.0 L/acre or Pretilachlor 50% EC
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PAGE 56: INTEGRATED WEED MANAGEMENT (IWM) & HERBICIDE ROTATION

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PAGE 57: INTEGRATED WEED MANAGEMENT (IWM) & HERBICIDE ROTATION

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PAGE 60: INTEGRATED WEED MANAGEMENT (IWM) & HERBICIDE ROTATION

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 - Post-Emergence Herbicides: Apply Bispyribac-sodium 10% SC @ 100 ml/acre in rice or Clodinafop-methyl 15% EC

PAGE 61: COMPREHENSIVE PRODUCTION PACKAGE FOR RICE (PADDY)

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule (NPK kg/ha): Apply balanced N:P₂O₅:K₂O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management (IPM) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 62: COMPREHENSIVE PRODUCTION PACKAGE FOR WHEAT

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \ (NPK kg/ha\): Apply balanced N:P₂O₅:K₂O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \ (IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 63: COMPREHENSIVE PRODUCTION PACKAGE FOR COTTON

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \ (NPK kg/ha\): Apply balanced N:P₂O₅:K₂O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \ (IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 64: COMPREHENSIVE PRODUCTION PACKAGE FOR TOMATO

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \ (NPK kg/ha\): Apply balanced N:P₂O₅:K₂O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \ (IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 65: COMPREHENSIVE PRODUCTION PACKAGE FOR MAIZE \(\text{CORN}\)

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(\text{NPK kg/ha}\): Apply balanced N:P₂O₅:K₂O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(\text{IPM}\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 66: COMPREHENSIVE PRODUCTION PACKAGE FOR POTATO

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \ (NPK kg/ha\): Apply balanced N:P₂O₅:K₂O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \ (IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 67: COMPREHENSIVE PRODUCTION PACKAGE FOR MUSTARD \(\text{RAPESEED}\)

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(\text{NPK kg/ha}\): Apply balanced N:P₂O₅:K₂O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(\text{IPM}\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 68: COMPREHENSIVE PRODUCTION PACKAGE FOR CHILLI \CAPSICUM\

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \NPK kg/ha\): Apply balanced N:P₂O₅:K₂O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \IPM\ with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 69: COMPREHENSIVE PRODUCTION PACKAGE FOR SUGARCANE

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \ (NPK kg/ha\): Apply balanced N:P₂O₅:K₂O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \ (IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 70: COMPREHENSIVE PRODUCTION PACKAGE FOR SOYBEAN

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \ (NPK kg/ha\): Apply balanced N:P₂O₅:K₂O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \ (IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 71: LIVESTOCK FARMING - DAIRY CATTLE & BUFFALO MANAGEMENT

Core Practices for Profitable Dairy:

1. Breeds & Genetics: High-yielding indigenous breeds (Gir, Sahiwal, Rathi) and buffal
2. Balanced Nutrition & TMR: Feed Total Mixed Ration (TMR) comprising 60% green fodder
3. Housing & Milking Hygiene: Free-stall loose housing with grooved non-slip concrete floor
4. Health & Mastitis Prevention: Practice post-milking teat dipping in 0.5% povidone-iodine

PAGE 72: LIVESTOCK FEED - FODDER PRODUCTION & SILAGE PRESERVATION

Year-Round Green Fodder Security:

1. High-Yielding Perennial Grasses: Super Napier \CO-4 / CO-5\ yielding 150-200 tonnes

PAGE 73: LIVESTOCK FARMING - SMALL RUMINANT MANAGEMENT \ (GOAT & SHEEP)
Commercial Goat & Sheep Production:

PAGE 74: LIVESTOCK FARMING - POULTRY PRODUCTION & BIOSECURITY
Poultry Systems & Management:

PAGE 75: INTEGRATED FARMING SYSTEM \IFS\ & RESOURCE RECYCLING
1-Hectare High-Income Integrated Farming System \IFS\:

PAGE 76: INTEGRATED PEST & DISEASE MANAGEMENT - IPM PRINCIPLES & ECONOMIC THRE
Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 77: INTEGRATED PEST & DISEASE MANAGEMENT - BIOLOGICAL PARASITOIDS & PREDATORS
Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release *Trichogramma* egg parasitoids @ 50,000/ha or spray *Beauveria*
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 78: INTEGRATED PEST & DISEASE MANAGEMENT - PHEROMONE & LIGHT TRAPPING SYSTEM
Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release *Trichogramma* egg parasitoids @ 50,000/ha or spray *Beauveria*
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 79: INTEGRATED PEST & DISEASE MANAGEMENT - SUCKING PEST COMPLEX MANAGEMENT Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 80: INTEGRATED PEST & DISEASE MANAGEMENT - LEPIDOPTERAN BORER MANAGEMENT

Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release *Trichogramma* egg parasitoids @ 50,000/ha or spray *Beauveria*
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 81: INTEGRATED PEST & DISEASE MANAGEMENT - MAJOR FUNGAL PATHOGEN CONTR
Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 82: INTEGRATED PEST & DISEASE MANAGEMENT - BACTERIAL DISEASES & BACTERIAL
Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release *Trichogramma* egg parasitoids @ 50,000/ha or spray *Beauveria*
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 83: INTEGRATED PEST & DISEASE MANAGEMENT - VIRAL VECTORS & BARRIER MANAG
Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 84: INTEGRATED PEST & DISEASE MANAGEMENT - ROOT-KNOT NEMATODE AMELIORA
Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 85: INTEGRATED PEST & DISEASE MANAGEMENT - FUNGICIDE RESISTANCE MANAGEMENT Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 86: FARM SAFETY & PRECAUTIONS - 10 COMMANDMENTS

Essential Farming Precautions & Safety Guidelines:

1. Chemical Spraying Precautions: Always wear Personal Protective Equipment \(\text{PPE}\) incl
2. Storage & Labeling: Store all pesticides in original labeled containers inside locked,

PAGE 87: FARMING EQUIPMENT & AGRICULTURAL MACHINERY

Modern Mechanization Solutions:

1. Primary Tillage Implements: Mouldboard (MB) Plough for deep inversion; Disc Plough
2. Secondary Tillage: Tractor Rotavator (PTO driven 540 RPM) preparing fine tilth in s
3. Precision Sowing & Planting Machinery: Multi-crop Pneumatic Planter ensuring singulated

PAGE 88: AGRICULTURAL SUSTAINABILITY & REGENERATIVE PRACTICES

Core Pillars of Sustainable Agriculture:

1. Soil Carbon Sequestration: Retaining crop residues in-situ increases soil organic carbo

PAGE 89: SEASONAL FARMING ACTIVITIES & MONTHLY CALENDAR

Comprehensive Month-by-Month Farming Guide:

- May - June \Pre-Kharif): Summer deep ploughing, cleaning irrigation channels, soil t

PAGE 90: RE-ENTRY INTERVAL \(\text{REI}\) & RESIDUE MANAGEMENT

Safe Agricultural Chemical Protocols:

- Re-Entry Interval \(\text{REI}\): The minimum time required between pesticide application and
- Pre-Harvest Interval \(\text{PHI}\): Mandatory duration between last spray and harvest: Manco
- Maximum Residue Limits \(\text{MRL}\): Strict compliance ensures domestic food safety and agr

PAGE 91: TRACTOR & MACHINERY OPERATIONAL SAFETY

Safety Protocols:

1. Tractor Rollover Protective Structures (ROPS) and seatbelts save lives in 99% of rollovers.
2. PTO Safety: Keep all Power Take-Off (PTO) safety guards intact; never step over rotating PTO shafts.
3. Combine Harvester: Disengage cutter bar clutch and shut off engine before clearing clog.
4. Electrical Safety: Maintain safe clearance (>3 meters) from overhead 11 kV high-tension power lines.

PAGE 92: EXTREME WEATHER PRECAUTIONS & LABOR SAFETY

1. Lightning Safety: Stop all field work during thunderstorms; avoid sheltering under isol
2. Heatstroke Prevention: Provide potable drinking water with electrolytes \\\(ORS\\) to fa
3. Cold Wave & Frost: Protect nursery beds with plastic tunnels or thatch coverings; irrig

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- Operational Impact: Boosts crop yields by 20-35%, saves 40-60% water, and reduces input

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PAGE 99: MODERN AGRICULTURE TECHNOLOGY & SCHEMES - PM-KUSUM SOLAR SCHEME &
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